INCH–POUND

MIL–DTL–211F
w/AMENDMENT 1
3 April 2008
SUPERSEDING
MIL–P–211F
24 July 2000

DETAIL SPECIFICATION

PHOSPHORUS, RED, TECHNICAL

This specification is approved for use by all Departments and Agencies of the Department of Defense.

1. SCOPE

1.1 Scope. This specification covers three classes of red phosphorus.

1.2 Classification. The classes of red phosphorus are specified as follows (see 6.2):

Class 1 – Dry
Class 3 – Dust suppressed with lubricating oil
Class 4 – Dust suppressed with aliphatic alcohol alkoxylate

2. APPLICABLE DOCUMENTS

2.1 General. The documents listed in this section are specified in sections 3 or 4 of this specification. This section does not include documents cited in other sections of this specification or recommended for additional information or as examples. While every effort has been made to insure the completeness of this list, document users are cautioned that they must meet all specified requirements of documents cited in sections 3 or 4 of this specification, whether or not they are listed.

Comments, suggestions, or questions on this document should be addressed to: U.S. Army Edgewood Chemical Biological Center, ATTN: AMSRD–ECB–ENA–S, 5183 Blackhawk Road, Aberdeen Proving Ground, MD 21010–5424 or emailed to SpecsTeam@apgea.army.mil. Since contact information can change, you may want to verify the currency of this address information using the ASSIST Online database at http://assist.daps.dla.mil.

AMSC N/A FSC 6810

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2.2 Government documents.

2.2.1 Specifications, standards, and handbooks. The following specifications, standards, and handbooks form a part of this document to the extent specified herein. Unless otherwise specified, the issues of these documents are those cited in the solicitation or contract.

FEDERAL STANDARDS

FEDERAL

FED−STD−313 − Material Safety Data, Transportation Data and Disposal Data for Hazardous Materials Furnished to the Government.

(Copies of these documents are available online at http://assist.daps.dla.mil/quicksearch/ or http://assist.daps.dla.mil or from the Standardization Document Order Desk, 700 Robbins Avenue, Building 4D, Philadelphia, PA 19111−5094.)

2.2.2 Other Government documents, drawings, and publications. The following other Government documents, drawings, and publications form a part of this document to the extent specified herein. Unless otherwise specified, the issues of these documents are those cited in the solicitation.

CODE OF FEDERAL REGULATIONS (CFR)

49 CFR 171 to 199 − Department of Transportation Hazardous Materials Regulations

(The Code of Federal Regulations is available online at www.access.gpo.gov or from the Superintendent of Documents, U.S. Government Printing Office, Washington, D.C. 20402. Orders for the above regulation should cite "49 CFR 171 to 199".)

2.3 Non−Government publications. The following documents form a part of this document to the extent specified herein. Unless otherwise specified, the issues of these documents are those cited in the solicitation or contract.

INTERNATIONAL CIVIL AVIATION ORGANIZATION

Technical Instructions for the Safe Transport of Dangerous Goods by Air

(Application for copies should be addressed to the International Civil Aviation Organization, 1000 Sherbrooke Street West, Suite 400, Montreal, Quebec, Canada H3A 2R2)

INTERNATIONAL MARITIME ORGANIZATION

International Maritime Dangerous Goods Code

(Application for copies should be addressed to the International Maritime Organization, 101−104 Piccadilly, London, WIV OAE, England)

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ASTM STANDARDS

B 527	– Standard Test Method for Determination of Tap Density of Metallic Powders and Compounds
D 92	– Flash and Fire Points by Cleveland Open Cup
D 97	– Pour Point of Petroleum Oils
D 130	– Detection of Copper Corrosion from Petroleum Products by the Copper Strip Tarnish Test
D 445	– Kinematic Viscosity of Transparent and Opaque Liquids (and the Calculation of Dynamic Viscosity)
D 974	– Acid and Base Number by Color−Indicator Titration
D 1193	– Reagent Water
D 1500	– ASTM Color of Petroleum Products (ASTM Color Scale)
D 2270	– Calculating Viscosity Index from Kinematic Viscosity at 40 and 100 °C
C 117	– Standard Test Method for Minerals Finer than 75-µm (No. 200) Sieve in Mineral Aggregates by Washing
E 11	– Wire−Cloth Sieves for Testing Purposes

(Copies of this document are available from www.astm.org or ASTM International, 100 Barr Harbor Drive, West Conshohocken, PA 19428−2959.)

2.4 Order of precedence. In the event of a conflict between the text of this document and the references cited herein, the text of this document takes precedence. Nothing in this document, however, supersedes applicable laws and regulations unless a specific exemption has been obtained.

3. REQUIREMENTS

3.1 Physical characteristics. Each class of red phosphorus shall conform to the applicable chemical and physical characteristics of table I when tested as specified therein. Class 3 red phosphorus shall be coated with lubricating oil conforming to the chemical and physical characteristics of table II when tested as specified therein. Class 4 hydrophilic dust suppressant shall be a straight chain, primary aliphatic alcohol alkoxylate added during manufacture prior to drying; the contractor shall certify that this requirement has been met (see 6.3).

3.2 Material Safety Data Sheets. Material Safety Data Sheets for red phosphorus shall be prepared and submitted by the contractor in accordance with FED−STD−313 (see 6.4).

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TABLE I. Chemical and physical characteristics

CHARACTERISTIC	PERCENT BY WEIGHT						Test Para.
	Class 1		Class 3		Class 4		
	Min	Max	Min	Max	Min	Max	
Red Phosphorus	99.0		97.0		97.0		4.2.4.3
White Phosphorus		0.02		0.02		0.02	4.2.4.1
Volatiles		0.25		0.25		0.25	4.2.4.6
Acidity (as H_3PO_4)		0.10		0.10		0.10	4.2.4.2
Dust Suppressant/Oil			0.8	1.3	0.8	1.5	4.2.4.4
Iron		0.06		0.06		0.06	4.2.4.3
Copper		0.005		0.005		0.005	4.2.4.3
Particle size:							4.2.4.5
Retained on No. 200 (75 µm) sieve	5	30	5	30	5	30	
Retained on No. 400 (38 µm) sieve	30	60	30	60	30	60	
Tap Density (g/cm^3)	1.2		1.2		1.2		4.2.4.7

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TABLE II. Lubricating oil chemical and physical characteristics

Characteristic	Requirement	ASTM test method
Viscosity, kinematic (centistokes) at 37.8°C	17 to 20	D 445
Flash point, minimum, °C	149	D 92
Pour point, maximum, °C	−34	D 97
ASTM Color, maximum	3	D 1500
Neutralization number, maximum	0.10	D 974
Copper corrosion, ASTM classification, maximum at 100°C	1	D 130
Viscosity index, minimum	30	D 2270

4. VERIFICATION

4.1 Responsibility for inspection. Unless otherwise specified in the contract or purchase order, the contractor is responsible for the performance of all inspection requirements (examinations and tests) as specified herein. Except as otherwise specified in the contract or purchase order, the contractor may use his own or any other facilities suitable for the performance of the inspection requirements specified herein, unless disapproved by the Government. The Government reserves the right to perform any of the inspections set forth in this specification where such inspections are deemed necessary to ensure supplies and services conform to prescribed requirements.

4.1.1 Responsibility for compliance. All items shall meet all requirements of sections 3 and 5. The inspection set forth in this specification shall become a part of the contractor's overall inspection system or quality program. The absence of any inspection requirements in the specification shall not relieve the contractor of the responsibility of ensuring that all products or supplies submitted to the Government for acceptance comply with all requirements of the contract. Sampling inspection, as part of manufacturing operations, is an acceptable practice to ascertain conformance to requirements, however, this does not authorize submission of known defective material, either indicated or actual, nor does it commit the Government to accept defective material.

4.1.2 Contractor assurance of compliance. The contractor's quality program or detailed inspection system shall provide assurance of compliance of all characteristics with the applicable

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drawing, special packaging instruction, and specification requirements using, as a minimum, the conformance criteria specified herein.

4.1.3 Alternative inspection provisions. Alternative inspection procedures, methods, or equipment, such as statistical process control, tool control, and other types of sampling procedures may be used by the contractor when they provide, as a minimum, the level or quality assurance required by the inspection provisions specified herein. Prior to applying such alternative procedures, methods, or equipment, the contractor shall describe them in a written proposal submitted to the Government for evaluation and approval. (See 6.9.) When required, the contractor shall demonstrate that the effectiveness of each proposed alternative is equal to or better than the quality assurance provisions specified herein. In cases of dispute as to whether the contractor's proposed alternative provides equal quality assurance, the provisions of these specification shall apply. All approved alternative inspection provisions shall be specifically incorporated into the contractor's quality program or detailed inspection system, as applicable.

4.2 Quality conformance inspection

4.2.1 Lotting. A lot shall consist of the red phosphorus of one class produced by one manufacturer, at one plant, from the same materials, and under essentially the same manufacturing conditions provided the operation is continuous. In the event the process is a batch operation, each batch shall constitute a lot (see 6.5).

4.2.2 Sampling.

4.2.2.1 For examination of packaging. Sampling shall be as specified in the contract (see 6.2).

4.2.2.2 For test (see 6.6). Sampling shall be conducted in accordance with table III. A representative specimen of 453 ± 3 grams (g) shall be removed from each sample container and placed in a suitable clean, dry container labeled to identify the lot and container from which it was taken.

TABLE III. Sampling for test

Number of containers in batch or lot	Number of sample containers
2 to 25	2
26 to 150	3
151 to 1,200	5
1,201 to 7,000	8
7,001 to 20,000	10
Over 20,000	20

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4.2.3 Inspection procedure.

4.2.3.1 For examination of packaging. The sample unit shall be one filled shipping container ready for shipment. Sample shipping containers shall be examined for the following defects:

(a) Contents per container not as specified
(b) Container not as specified
(c) Container closure not as specified
(d) Container damaged or leaking
(e) Marking incorrect, missing, or illegible
(f) Bag exterior or drum interior or exterior contaminated with red phosphorus

4.2.3.2 For test (see 6.6). Approximately equal portions of all of the specimens taken in 4.2.2.2 shall be thoroughly mixed to form a composite specimen of no less than 450 g. The composite specimen shall be tested as specified in the applicable portions of 4.2.4. Each test shall be conducted in duplicate analysis. Failure of either analysis of any test shall be cause for rejection of the lot represented.

4.2.4 Tests (see 6.6)

(a) Inductive Coupled Plasma (ICP). Be sure all glassware is free of metals. Soak glassware in 5% HNO_3, rinse with distilled water, soak in 5% HCl and rinse with distilled water. Run ICP according to manufacturers specification. Assure the ICP has been purged with nitrogen at 223.008 nm for Cu, 238.204 nm for Fe, and 177.494 nm for P. Calibrate the instrument using standards of Cu, Fe, and P. Run a water blank before running samples.

(b) Chemicals. Water, in accordance with ASTM D 1193, and reagent grade chemicals shall be used throughout the tests. Where applicable, blank determinations and standards shall be run and corrections applied where significant. ICP standards for copper and iron shall be 5 µg/ml and standards for phosphorus shall be 50 µg/ml. Tests shall be conducted as follows:

4.2.4.1 White Phosphorus. Weigh 10 g of the RP specimen and add 50 milliliters (ml) carbon disulfide. Mix for 60 minutes in a sonic bath or with magnetic stirrer. Filter through a Whatman No. 42 filter paper or equal and take a 10 ml aliquot into a 50 ml volumetric flask and allow to evaporate dry in hood with air or nitrogen stream. Add 25 ml water, 2.5 ml concentrated nitric acid and dilute to volume with water. Run ICP on prepared solution.

$$\text{Percent White Phosphorus} = \frac{25 \times 10^{-3} A}{W}$$

where: A = ICP reading (µg/ml)
W = weight of specimen (g)

4.2.4.2 Acidity as Phosphoric Acid. Weigh 10 g of the RP specimen in a 50 ml flask and dilute to volume with water. Mix in sonic bath for 5 minutes. Filter through a Whatman No. 42 filter paper or equal and run ICP on solution.

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$$\text{Percent Acidity as Phosphoric Acid} = \frac{15.8 \times 10^{-3} A}{W}$$

where: A = ICP reading (µg/ml)
W = weight of specimen (g)

4.2.4.3 Red Phosphorus, Copper, Iron.

(a) Sample in solution. Weigh 0.2±.05 g (to nearest 0.1 mg) of the RP into a 400 ml flask. Add 50 ml of distilled water, 25 ml of concentrated nitric acid, and 25 ml of concentrated hydrochloric acid. Heat the sample slowly until there is approximately 30 ml. Remove the sample from the source of heat and cover with a watch glass if the reaction becomes violent. If particulates are still present, add 10 ml of nitric acid and wash down the inside of the beaker with distilled water. Evaporate the contents to approximately 30 ml, cool and filter (with a Whatman 42 or equal) into a 100 ml volumetric flask. Dilute to volume with water.

(b) Copper. Run prepared solution in ICP

$$\text{Percent Cu} = \frac{.01A}{W}$$

where: A = concentration of Cu from ICP in µg/ml
W = weight of sample in grams

(c) Iron. Run prepared solution in ICP

$$\text{Percent Fe} = \frac{.01A}{W}$$

where: A = concentration of Fe from ICP in µg/ml
W = weight of sample in grams

(d) Red Phosphorus. Take 5 ml sample from 100 ml flask and dilute to 200 ml with distilled water. Run solution in ICP. Blanks are also run to determine if adjustments are necessary.

$$\text{Percent RP} = \left(\frac{.4A}{W}\right) - E - 0.316F$$

where: A = concentration of P from ICP, in µg/ml
W = weight of sample, in grams
E = Percent WP from 4.2.4.1
F = Percent H_3PO_4 from 4.2.4.2
* See note 6.10 if instrument drift is present.

4.2.4.4 Dust Suppressant. Weigh 5.0 g of specimen into a 150 ml beaker and add 50 ml of methylene chloride and a 1 inch stirring bar. Cover with a watch glass and stir for 30 minutes. Filter the solution through a Whatman No. 42 or equal filter paper into a clean, tared, glass

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evaporating dish. Wash the beaker and paper three times into the evaporating dish using a total of 50 ml of methylene chloride. Place the evaporating dish containing the filtrate and washings on a steam bath and evaporate to about 3 ml of solution. Do not heat the residue to dryness as loss of dust suppressant may occur. Remove from the steam bath and evaporate the remaining methylene chloride in an air or nitrogen stream. Dry off traces of outside moisture, cool in a desiccator, and weigh. Calculate the percent by weight dust suppressant as follows:

$$\text{Percent dust suppressant} = 100(A-B)/W$$

where: A = Weight of dish and residue, in grams
B = Weight of dish, in grams
W = Weight of specimen, in grams

4.2.4.5 Particle Size. (NOTE: Use ASTM C117 as a guideline in conducting this analysis) Tare sieves conforming to ASTM E11. Both sieves shall be 3 inches in diameter. Mix 10 to 13 grams of the specimen with 100 ml of water. Pour RP/water mixture into the sieve and wash the specimen through with more water until the quantity of material passing through the sieve does not exceed a trace. Sieving may be conducted using one sieve at a time to aid in the sieving process. Assure all water and specimen washed through first sieve is retained for subsequent sieve. Dry the sieves in a laminar flow bench for 7 to 8 hours. Calculate the percent of specimen retained as follows:

$$\text{Percent retained} = \frac{100\,(A-B)}{W}$$

where: A = Weight of the sieve and contents, after drying, in grams
B = Tare weight of the sieve, in grams
W = Weight of specimen, in grams

Percent retained on the second sieve is added to the first sieve.

4.2.4.6 Volatiles. Tare a clean, dry watch glass and record all weights to the nearest 0.1 mg. Add 5 g of the RP specimen to the watch glass and record the gross weight. Place the watch glass and specimen in a drying oven at 103 °C for 2 hours. At the end of the drying time, place the watch glass and specimen in a desiccator. Cool and weigh to a constant weight. Calculate the percent by weight volatiles as follows:

$$\text{Percent volatiles} = 100\,(A-B)/W$$

where: A = Weight of specimen and container before drying, in grams
B = Weight of specimen and container after drying, in grams
W = Weight of specimen, in grams

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4.2.4.7 Tap Density

(a) Procedure. The procedures of ASTM B 527 shall be followed with the following exceptions:

(1) Tapping apparatus shall be equipped with a 100 ml graduated cylinder
(2) Total number of taps shall be set to 1500 counts
(3) Sample size shall be 50 grams

5. PACKAGING

5.1 Packaging. For acquisition purposes, the packaging requirements shall be as specified in the contract or order (see 6.2). When packaging of materiel is to be performed by DoD or in−house contractor personnel, these personnel need to contact the responsible packaging activity to ascertain packaging requirements. Packaging requirements are maintained by the Inventory Control Point's packaging activities within the Military Service or Defense Agency, or within the military service's system commands. Packaging data retrieval is available from the managing Military Department's or Defense Agency's automated packaging files, CD−ROM products, or by contacting the responsible packaging activity.

6. NOTES

(This section contains information of a general or explanatory nature that may be helpful, but is not mandatory.)

6.1 Intended use. Red phosphorus is intended for use in military pyrotechnic mixtures. Use of class 3 and 4 red phosphorus will reduce dust during pyrotechnic processing.

6.2 Acquisition requirements. Acquisition documents must specify the following:

(a) Title, number, and date of this specification
(b) DELETED
(c) Class of red phosphorus required (see 1.2)
(d) Packaging required (see SPI PMIL−DTL−211)

6.3 Aliphatic alcohol alkoxylate dust suppressant. Alkasurf LA−EP 25LF, manufactured by Alkaril Chemicals LTD, has been found to meet the material requirements.

6.4 Material Safety Data Sheets. Contracting officers will identify those activities requiring copies of completed Material Safety Data Sheets prepared in accordance with FED−STD−313. The pertinent mailing addresses for submission of data are listed in FED−STD−313.

6.5 Batch. A batch is defined as that quantity of material which has been manufactured by some unit chemical process or subjected to some physical mixing operation intended to make the final product substantially uniform.

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6.6 Sampling and testing precaution. This specification may involve hazardous materials, operations, and equipment. This specification does not purport to address all of the safety problems associated with its use. It is the responsibility of the user of this specification to establish appropriate safety and health practices and determine the applicability of regulatory limitations prior to use.

6.7 Interplant shipment. DELETED

6.8 Significant places. For the purpose of determining conformance with this specification, an observed or calculated value should be rounded off 'to the nearest unit' in the right−hand place of figures used in expressing the limiting value, in accordance with the rounding−off method of ASTM E 29.

6.9 Submission of alternative inspection provisions. Proposed alternative inspection provisions should be submitted by the contractor to the procuring contracting officer for evaluation and approval by the technical activity responsible for preparation of this specification.

6.10 Inductive Coupled Plasma drift. Standards run at regular intervals during the ICP tests will indicate any drifts in the ICP instrument. If drift is apparent, correct the series of values obtained between each Standard run as follows:

$$\text{Corrected Value} = (S_T/(S_B + S_A)/2) \times V$$

where: S_T = Theoretical standard value
S_B = Standard value before series
S_A = Standard value after series
V = Measured value within series

6.11 Changes from previous issue. DELETED.

6.12 Supersession data. DOD−P−51463(EA), dated 12 February 1980, with Amendment 1, dated 9 June 1982, and Notice 1, dated 1 June 1989, is superseded by the class 3 red phosphorus of this specification. Future acquisition of the oiled red phosphorus formerly acquired under DOD−P−51463(EA) should be made by specifying class 3 of MIL−P−211.

6.13 Subject term (key word) listing.

Absolute density
Acidity
Dust suppressant
Particle size
Volatile matter
White phosphorus

6.14 Amendment notations. The margins of this specification are marked with vertical lines to indicate modifications generated by this amendment. This was done as a convenience only and the Government assumes no liability whatsoever for any inaccuracies in these notations. Bidders

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